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Assessment & Course Details:

Course Code: XBCS2183N		**Course Name: Operating System**	
Program Name: Bachelor of Computer Science (Hons)			
Lecturer/s Name: Ts. Aidora Abdullah			
Assessment Due Date: *(dd/mm/yy)*	07 / 08 / 2026	**Assessment Title:**	Individual Assignment Report

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1.0 Introduction

This assignment will reflect Operating System memory management under virtual memory limit of 256MB. This limits the per process swapping memory allocation, which means that the OS must carefully balance the swapping process between RAM and disk. This constraint highlights how memory management strategies affect performance, multitasking, and stability.

Core concepts:

- Virtual memory: Extends RAM by using disk storage to store inactive pages
- Paging: Process of dividing memory into fixed-size blocks and swapping between RAM and disk.
- Thrashing: Performance degradation due to excessive paging.

2.0 Research Question:

How does different load with 256MB virtual memory cap affect the stability, swap pattern, as well as the time spent in kernel space and user space?

3.0 Test cases:

Ulimit = 256MB

Stressing tool: stress-ng

Monitoring tool: vmstat

No	Description	No. of Workers	Total allocation
1	Many workers without exceeding the VM limit	2	2 x 128MB = 256MB
2	Many workers exceeding the VM limit	2	2 x 180MB = 360MB

3	1 worker without exceeding the VM limit	1	1 x 256MB = 256MB

4.0 Runtime evidence

```
[2026-08-11 21:11:46] procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
[2026-08-11 21:11:46] r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
[2026-08-11 21:11:46] 3  0      0 1078112   4532 1014544    0    0  1392   166 2526    4  2  4 93  0  0  0
[2026-08-11 21:11:47] 4  0  25756 917644   4532 1035484    0 25644  4032 27248 2365 1186 40 57  3  0  0  0
[2026-08-11 21:11:49] 3  0    528 848036   4532 1035692    0 105428     0 108960 3330  803 49 51  0  0  0  0
[2026-08-11 21:11:49] 4  0    544 862208   4532 1035792    0 131072     0 131155 4104 2221 62 38  0  0  0  0
[2026-08-11 21:11:50] 2  0    544 818076   4532 1035788    0    0     0     0 3511 1260 71 29  0  0  0  0
[2026-08-11 21:11:51] 2  0    544 895888   4532 1035820    0    0     0     0 3394  394 91  9  0  0  0  0
[2026-08-11 21:11:53] 3  0    544 848404   4532 1035876    0    0     0     8 3439  405 93  7  0  0  0  0
[2026-08-11 21:11:53] 2  0    544 879776   4532 1035928    0    0     0     0 4098 2511 82 18  0  0  0  0
[2026-08-11 21:11:54] 3  0    544 889548   4532 1036000    0    0     8     0 3482  676 89 11  0  0  0  0
[2026-08-11 21:11:56] 2  0    412 930600   4532 1036080    0 65536     0 65536 4013 2590 60 40  0  0  0  0
[2026-08-11 21:11:57] 2  0    412 1001368   4532 1036152    0    0     0     0 3439  658 90 10  0  0  0  0
[2026-08-11 21:11:58] 2  0    412 1007476   4532 1036192    0    0     0     0 3361  440 95  5  0  0  0  0
[2026-08-11 21:11:59] 2  0    412 1012904   4532 1036232    0    0     0     4 3307  347 96  4  0  0  0  0
[2026-08-11 21:12:00] 4  0    412 1016464   4532 1036256    0    0     0     4 3308  433 94  6  0  0  0  0
[2026-08-11 21:12:01] 2  0    412 1020960   4532 1036296    0    0     0     0 3219  365 95  5  0  0  0  0
[2026-08-11 21:12:02] 2  0    412 1023096   4532 1036336    0    0     0     0 3313  349 95  5  0  0  0  0
[2026-08-11 21:12:03] 2  0    412 1023072   4532 1036368    0    0     0     0 3338  367 95  5  0  0  0  0
[2026-08-11 21:12:04] 2  0    412 1022572   4532 1036408    0    0     0     0 3241  356 95  5  0  0  0  0
[2026-08-11 21:12:05] 2  0    412 1022572   4532 1036464    0    0     0     4 3272  371 95  5  0  0  0  0
[2026-08-11 21:12:06] 2  0    412 1022572   4532 1036536    0    0     0     0 3430  393 95  5  0  0  0  0
[2026-08-11 21:12:07] 2  0    412 1022572   4532 1036592    0    0     0     0 3437  417 95  5  0  0  0  0
[2026-08-11 21:12:08] procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
[2026-08-11 21:12:08] r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
[2026-08-11 21:12:08] 3  0    412 1022572   4532 1036660    0    0     0     0 3437  422 95  5  0  0  0  0
[2026-08-11 21:12:09] 2  0    412 1022572   4532 1036692    0    0     0     0 3318  354 95  5  0  0  0  0
[2026-08-11 21:12:10] 2  0    412 1022572   4532 1036764    0    0     0     0 3395  454 95  5  0  0  0  0
[2026-08-11 21:12:11] 2  0    412 1022572   4532 1036804    0    0     0     4 3413  381 94  6  0  0  0  0
[2026-08-11 21:12:12] 2  0    412 1022572   4532 1036876    0    0     0     0 3338  378 94  6  0  0  0  0
[2026-08-11 21:12:13] 4  0    412 1022572   4532 1036916    0    0     0     0 3988 3002 85 15  0  0  0  0
[2026-08-11 21:12:14] 4  0    412 1022324   4532 1036988    0    0     0     0 3533  847 93  7  0  0  0  0
[2026-08-11 21:12:15] 2  0    412 1022072   4532 1037044    0    0     0     0 3562  833 92  8  0  0  0  0
[2026-08-11 21:12:16] 2  0    412 1021064   4532 1037084    0    0     0     0 3430  553 93  7  0  0  0  0
```

Figure 4.1: vmstat log for test case 1

```
[2026-08-11 21:13:58] procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
[2026-08-11 21:13:58] r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
[2026-08-11 21:13:58] 4  0    668 1069820   4532 1026808    0  404  1177   560 2576    5  5  5 90  0  0  0
[2026-08-11 21:13:59] 2  0   1244 838104   4532 1043672    0 184320     0 184320 4455 3500 34 66  0  0  0
[2026-08-11 21:14:00] 3  0   1244 725876   4532 1043532    0    0     0     0 1349  483 42 58  0  0  0  0
[2026-08-11 21:14:01] 2  0   1444 957484   4532 1043652    0 92160     0 92160 3535  935 44 31  0 25  0  0
[2026-08-11 21:14:02] 2  0   1444 759140   4532 1043660    0    0     0     0 3399  350 73 27  0  0  0  0
[2026-08-11 21:14:03] 3  0   1204 810612   4532 1043692    0 92160     0 92160 3543  704 64 31  0  5  0  0
[2026-08-11 21:14:04] 2  0   1204 694128   4532 1043692    0    0     0     8 3427  540 85 15  0  0  0  0
[2026-08-11 21:14:05] 2  0   1204 722052   4532 1043716    0    0     0     0 3429  442 92  8  0  0  0  0
[2026-08-11 21:14:06] 2  1   1204 793304   4532 1051140    0    0  7680     0 3670 1247 82 18  0  0  0  0
[2026-08-11 21:14:07] 5  0   1204 749508   4532 1073068    0    0 22060   126 3981 1949 81 19  0  0  0  0
[2026-08-11 21:14:08] 2  0   1204 677416   4532 1073296    0    0     0     0 3690 1751 86 14  0  0  0  0
[2026-08-11 21:14:09] 2  0   1204 669324   4532 1073508    0    0     0     0 3597  509 92  8  0  0  0  0
[2026-08-11 21:14:10] 5  0   1204 737352   4532 1073716    0    0     0     4 3945 1719 80 20  0  0  0  0
[2026-08-11 21:14:11] 2  0   1204 742800   4532 1073912    0    0     0   100 3779 1223 82 18  0  0  0  0
[2026-08-11 21:14:12] 2  0   1204 820552   4532 1074116    0    0     0     0 3950 1760 79 21  0  0  0  0
[2026-08-11 21:14:13] 2  0   1204 820452   4532 1074316    0    0     0     0 3820 1312 89 11  0  0  0  0
[2026-08-11 21:14:14] 3  0   1204 849520   4532 1074612    0    0     0     0 4294 2263 83 17  0  0  0  0
[2026-08-11 21:14:15] 3  0   1204 849116   4532 1074716    0    0     0     0 4216 2098 86 14  0  0  0  0
[2026-08-11 21:14:16] 5  0   1204 879048   4532 1075060    0    0     0     4 4136 1959 83 17  0  0  0  0
[2026-08-11 21:14:17] 2  0   1204 909304   4532 1075396    0    0     0     0 4558 2800 82 18  0  0  0  0
[2026-08-11 21:14:18] 2  0   1204 907980   4532 1075596    0    0     0     0 4057 1733 89 11  0  0  0  0
[2026-08-11 21:14:19] procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
[2026-08-11 21:14:19] r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
[2026-08-11 21:14:20] 5  0   1204 920400   4532 1075964    0    0     0  2249 4245 2706 82 18  0  0  0  0
[2026-08-11 21:14:20] 2  0   1204 920200   4532 1076136    0    0     0   964 3945 1268 91  9  0  0  0  0
[2026-08-11 21:14:21] 2  0   1204 920200   4532 1076136    0    0     0  1688 3839 1071 92  8  0  0  0  0
[2026-08-11 21:14:22] 2  0   1204 920200   4532 1076276    0    0     0     4 3769 1338 92  8  0  0  0  0
[2026-08-11 21:14:23] 2  0   1204 920200   4532 1076396    0    0     0     0 3888 1475 91  9  0  0  0  0
[2026-08-11 21:14:24] 2  0   1204 919952   4532 1076476    0    0     0     0 3519  957 92  8  0  0  0  0
[2026-08-11 21:14:25] 2  0   1204 919952   4532 1076596    0    0     0     0 3753 1668 89 11  0  0  0  0
[2026-08-11 21:14:26] 2  0   1204 919952   4532 1076660    0    0     0     0 3911 2207 90 10  0  0  0  0
[2026-08-11 21:14:27] 2  0   1204 919952   4532 1076708    0    0     0     0 3562 1292 93  7  0  0  0  0
```

Figure 4.2: vmstat log for test case 2

```
[2026-08-11 21:57:14] procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
[2026-08-11 21:57:14] r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
[2026-08-11 21:57:14] 3  0    436 229276   4532 1445904    0  205   318   383 2395    3  4  4 92  0  0  0
[2026-08-11 21:57:15] 0  0    436 205652   4532 1462604    0    0     0     0 3288 2248  6 12 82  0  0  0
[2026-08-11 21:57:16] 0  0    436 205580   4532 1462904    0    0     0     0 2386  532  1  5 95  0  0  0
[2026-08-11 21:57:17] 0  0    436 205580   4532 1463064    0    0     0     0 3165 2428  5 11 84  0  0  0
[2026-08-11 21:57:18] 0  0    436 205580   4532 1463280    0    0     0     0 2659 1094  2  7 92  0  0  0
[2026-08-11 21:57:19] 0  0    436 205332   4532 1463444    0    0     0     0 2420  579  1  5 95  0  0  0
[2026-08-11 21:57:20] 2  0    436 205332   4532 1463636    0    0     0    12 2415  539  1  4 96  0  0  0
[2026-08-11 21:57:21] 0  0    436 205080   4532 1463756    0    0     0     0 2346  523  1  5 95  0  0  0
[2026-08-11 21:57:22] 0  0    436 204832   4532 1463948    0    0     0     0 2394  528  1  4 95  0  0  0
[2026-08-11 21:57:23] 0  0    436 204584   4532 1464192    0    0     0     0 2412  579  1  6 94  0  0  0
[2026-08-11 21:57:24] 1  0    436 204584   4532 1447916    0    0     0     0 2471  701  1  5 94  0  0  0
[2026-08-11 21:57:25] 0  0    436 204584   4532 1447980    0    0     0     0 2412  658  1  4 95  0  0  0
[2026-08-11 21:57:26] 0  0    436 204584   4532 1448252    0    0     0     8 2274  504  0  4 96  0  0  0
[2026-08-11 21:57:27] 0  0    436 204584   4532 1448436    0    0     0     0 2325  506  1  4 96  0  0  0
[2026-08-11 21:57:28] 0  0    436 204336   4532 1448600    0    0     0     0 2432  501  1  5 94  0  0  0
[2026-08-11 21:57:29] 0  0    436 204088   4532 1448948    0    0     0     0 2363  493  1  6 94  0  0  0
[2026-08-11 21:57:30] 0  0    436 204088   4532 1449100    0    0     0     0 2259  460  1  4 95  0  0  0
[2026-08-11 21:57:31] 1  0    436 204088   4532 1449240    0    0     0     0 2300  539  0  3 97  0  0  0
[2026-08-11 21:57:32] 0  0    436 204088   4532 1449476    0    0     0     4 2315  534  1  4 95  0  0  0
[2026-08-11 21:57:33] 0  0    436 204088   4532 1449700    0    0     0     0 2381  539  1  5 95  0  0  0
[2026-08-11 21:57:34] 0  0    436 203840   4532 1449924    0    0     0     0 2382  458  1  4 95  0  0  0
[2026-08-11 21:57:35] procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
[2026-08-11 21:57:35] r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
[2026-08-11 21:57:35] 1  0    436 203840   4532 1450052    0    0     0     0 2330  479  1  4 95  0  0  0
[2026-08-11 21:57:36] 1  0    436 203840   4532 1450204    0    0     0     0 2311  466  0  4 96  0  0  0
[2026-08-11 21:57:37] 0  0    436 203592   4532 1450364    0    0     0     0 2226  485  0  4 96  0  0  0
[2026-08-11 21:57:38] 1  0    436 203592   4532 1450512    0    0     0    22 2309  485  1  4 96  0  0  0
[2026-08-11 21:57:39] 0  0    436 203592   4532 1450688    0    0     0  2464 2331  454  1  4 95  0  0  0
[2026-08-11 21:57:40] 0  0    436 203592   4532 1450844    0    0     0     0 2336  444  1  4 95  0  0  0
[2026-08-11 21:57:41] 0  0    436 203592   4532 1451008    0    0     0     0 2368  476  0  4 96  0  0  0
[2026-08-11 21:57:42] 1  0    436 203344   4532 1451256    0    0     0     0 2410  480  1  4 95  0  0  0
[2026-08-11 21:57:43] 0  0    436 203344   4532 1451408    0    0     0     0 2352  502  1  4 95  0  0  0
```

Figure 4.3: vmstat log for test case 3

5.0 Analysis and Findings

<u>2x 128MB:</u> **Swap Pattern:** Minor startup swap • Minor swap-out at startup as the kernel evicts inactive pages. After that, swap activity drops to zero. • The worker requests exactly 256 MB, which matches the ulimit. **CPU behaviour:** High User CPU • CPU time spent on user space is high (> 80%) because both workers actively touch memory. • System CPU rises moderately due to context switching (cs) and page table updates. • Idle drops to 10–20%, this shows that the CPU doesn't spend more time waiting. **Stability:** Stable • Both workers run smoothly, no sustained swap and memory access stays in RAM which avoids disk bottlenecks. • This reflects the utilization of the cap, which means that it is just right for the stress-ng workers to run without oversubscription.
<u>2x 180MB:</u> **Swap Pattern:** Swap burst • Continuous swap bursts (so=184,320 KB/s, so=92,160 KB/s). Pages are evicted repeatedly to enforce the cap. **CPU behaviour:** High System CPU • System CPU (sys) spikes (> 30%) as the kernel handles page faults and swap I/O. • User CPU (us) drops during bursts, because workers stall waiting for memory. • Idle collapses, and I/O wait (wa) rises (up to 25%), showing the disk bottleneck.

Stability: Degraded (Thrashing) • Thrashing occurred because the workload exceed the ulimit of 256MB. It spends more time on swapping pages out the disk instead of executing the process instructions.
<u>1x 256MB:</u> **Swap Pattern:** Brief startup swap • Kernel evicts a few inactive cache pages (so bursts) to make contiguous space. After that, no sustained swap occurs. • The worker requests exactly 256 MB, which matches the ulimit. **CPU behaviour:** Mostly Idle • Time spent in user space (us) and kernel space (sys) is low since idle (id) stays high (> 90%) because one worker does not saturate cores. • User CPU (us) is modest, because one worker touches the memory sequentially without saturating cores. • System CPU (sys) low, because page faults are minimal. **Stability:** Very stable • No thrashing and predictable performance. • Kernel enforces cap efficiently. • This case demonstrates how Linux handles an exact-fit allocation with fairness and stability.

6.0 Script Design

```
#!/bin/bash

#Dihrran Chong Yun Siong
#0139028
#Investigation Area:  Memory Management
#Workload Type: Memory-intensive
#Experimental variable: ulimit 256MB
#Observation Source: vmstat

while true
do
# testing different workload with different numbers of workers to see the difference
echo "1. Start Testing 2x 128MB"
echo "2. Start Testing 2x 180MB"
echo "3. Start Testing 1x 256MB"
echo "4. View vmstat.log"
echo "5. View workload.log"
echo "99. Exit"

read -p "Enter choice: " choice

case $choice in
```

Figure 6.1: Script headers and choices for test cases

```
1)

echo "---experiment for 2x 128MB load---"
echo "---APPLYNG ULIMIT 256MB---"
ulimit -v 262144

echo "---STARTING VMSTAT MONITORING---"
vmstat 1 30 | while read line; do
# prints the timestamp to observe the behavior as time goes
echo "$(date '+[%Y-%m-%d %H:%M:%S]') $line"
#save the output to vmstat.log and run in the background
done > vmstat.log &
VMSTAT_PID=$!

# --- Run workload --- first test by generating 128mb x 2 of allocation (just enough for the vm)
echo "---RUNNING MEMORY INTENSIVE WORKLOAD---"
stress-ng --vm 2 --vm-bytes 128M --timeout 30s 2>&1 | while read line; do
echo "$(date '+[%Y-%m-%d %H:%M:%S]') $line"
done > workload.log

# --- Cleanup ---
wait $VMSTAT_PID
echo "EXPERIMENT COMPLETED"
echo "Logs saved in vmstat.log for monitoring and in workload.log for workload"
;;
```

Figure 6.2: test case 1

```
2)
echo "---experiment for 2x 180MB load---"
echo "---APPLYNG ULIMIT 256MB---"
ulimit -v 262144

echo "---STARTING VMSTAT MONITORING---"
vmstat 1 30 | while read line; do
echo "$(date '+[%Y-%m-%d %H:%M:%S]') $line"
done > vmstat.log &
VMSTAT_PID=$!

# --- Run workload --- Next test by allocate beyond vm memory limit to see how bad the result
echo "---RUNNING MEMORY INTENSIVE WORKLOAD---"
stress-ng --vm 2 --vm-bytes 180M --timeout 30s 2>&1 | while read line; do
echo "$(date '+[%Y-%m-%d %H:%M:%S]') $line"
done > workload.log

# --- Cleanup ---
wait $VMSTAT_PID
echo "EXPERIMENT COMPLETED"
echo "Logs saved in vmstat.log for monitoring and in workload.log for workload"
;;
```

Figure 6.3: test case 2

```
3)
echo "---experiment for 1x 256MB load---"
echo "---APPLYNG ULIMIT 256MB---"
ulimit -v 262144

echo "---STARTING VMSTAT MONITORING---"
vmstat 1 30 | while read line; do
echo "$(date '+[%Y-%m-%d %H:%M:%S]') $line"
done > vmstat.log &
VMSTAT_PID=$!

# --- Run workload --- lastly applying just enough load but use only1 worker
echo "---RUNNING MEMORY INTENSIVE WORKLOAD---"
stress-ng --vm 1 --vm-bytes 256M --timeout 30s 2>&1 | while read line; do
echo "$(date '+[%Y-%m-%d %H:%M:%S]') $line"
done > workload.log

# --- Cleanup ---
wait $VMSTAT_PID
echo "EXPERIMENT COMPLETED"
echo "Logs saved in vmstat.log for monitoring and in workload.log for workload"
;;

4)
cat vmstat.log
;;

5)
cat workload.log
;;

99)
exit 0
;;

*)
echo "Invalid option"
;;

esac

done
```

Figure 6.4: test case 3